Edge Detection (Basic Idea)

An **edge** is where **pixel brightness changes suddenly**.

Example image:

White area | Black object

Pixel values:

255 255 255
255 255 255
0 0 0

Between **255 → 0** there is a **big change**.

That location is an **edge**.

So edge detection means:

Find where pixel values change sharply.

Canny Edge Detector

Canny is the **most popular edge detection method**.

Steps

1️⃣ Smooth image (remove noise)
2️⃣ Detect intensity change
3️⃣ Keep thin edges
4️⃣ Remove weak edges

Example:

Original pixels:

255 255 255
255 255 255
0 0 0

Edge detected:

0 0 1
0 0 1
0 0 1

1 = edge

Why good?

✓ clean edges
✓ less noise

Example use:

- road lane detection
- object boundaries

LoG (Laplacian of Gaussian)

Full form:

Laplacian of Gaussian

Idea:

1️⃣ Smooth image using **Gaussian blur**
2️⃣ Use **Laplacian filter** to detect rapid change.

Example pixels:

255 255 255
200 200 200
50 50 50

Large change between **200 → 50** → edge.

So LoG finds edges by checking **second-order intensity change**.

Used for:

- fine edge detection
- blob detection

DoG (Difference of Gaussian)

DoG is simpler than LoG.

Steps:

1️⃣ Blur image with small Gaussian
2️⃣ Blur image with larger Gaussian
3️⃣ Subtract them

Example:

First blur:

255 254 255
255 254 255
255 254 255

Second blur:

254 254 254
254 254 254
254 254 254

Subtract:

1 0 1
1 0 1
1 0 1

Edges appear.

So:

DoG detects edges by subtracting two blurred images.

Used in:

- SIFT feature detection

Line Detection – Hough Transform

After edge detection we get **many edge pixels**.

Example edge image:

0 0 1 0
0 0 1 0
0 0 1 0
0 0 1 0

These pixels form a **vertical line**.

Hough Transform finds that line.

Idea:

Each edge pixel **votes for possible lines**.

The line with **maximum votes** is detected.

Used in:

- lane detection
- document borders
- road detection

Corner Detection

A **corner** is where **two edges meet**.

Example:

```
+-----+
|    |
|    |
+-----+
```

Corners:

● ●
● ●

Corners are useful for:

- matching images
- tracking objects
- 3D reconstruction

Harris Corner Detector

Harris checks **pixel changes in two directions**.

Example pixel window:

255 255 255
255 0 0
255 0 0

Intensity changes:

- horizontally
- vertically

So it marks this point as a **corner**.

Hessian Affine Detector

Hessian detector finds **important points (blobs / corners)** that stay stable even if:

- image zoom changes
- image rotates
- scale changes

Used in:

- feature matching
- object recognition